

FIRE: Fisher Information Recommendation Engine

Psychometric Content Routing for Recommendation Systems

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Abstract

We introduce FIRE (Fisher Information Recommendation Engine), a recommendation algorithm grounded in Item Response Theory (IRT) and the Linear Logistic Test Model (LLTM). FIRE operationalizes the PCR frontier value, $P(1 - P)$, as a Fisher-information-shaped exploration term; for warm IRT-scored items this reduces to the standard local Fisher information criterion used in computerized adaptive testing. FIRE combines (1) IRT-based engagement prediction for exploitation, (2) normalized Fisher Information for exploration, and (3) an LLTM item-feature-based difficulty prior for low-count items. On MovieLens-1M [Harper and Konstan, 2015] (6,040 users, 3,883 valid movies), with full-catalog candidate evaluation over 300 randomly sampled eligible users per seed across 3 seeds, FIRE achieves AUC 0.7682 ± 0.0002 and provides $1.8\times$ more catalog coverage than IRT among learned models (0.0179 vs 0.0100), with explicit per-component score decompositions. LLTM achieves a squared correlation of $\rho^2 = 0.564$ between 22 observable item features and fitted warm-item difficulty, in-sample. On low-count items, LLTM interpolation did not improve prediction over IRT, indicating that FIRE’s current empirical advantage is catalog coverage rather than cold-start accuracy. In the included implementation, the core model completes in approximately 97 seconds per seed in local CPU runs; runtime may vary with hardware and numerical-library threading.

1 Introduction

Social media recommendation systems share a multi-objective scoring structure. A leaked internal TikTok document describes a simplified value-weighted formula based on predicted interactions [Business Insider, 2021]; we use this as an architectural analogy only, not as an official platform specification. Similar scoring underlies YouTube’s two-stage pipeline [Covington et al., 2016] and Meta’s content systems [Meta AI, 2023].

FIRE is motivated by cold-start, diversity, and transparency challenges [Kaminskas and Bridge, 2016, European Union, 2024], but in this MovieLens-1M experiment its demonstrated empirical advantage is catalog coverage among learned models, not cold-start accuracy. FIRE builds on psychometric theory: the Frontier Value $F = P(1 - P)$ from PCR [Kang, 2026] is mathematically equivalent to Fisher Information [Lord and Novick, 1968]—the criterion BanditCAT [Sharpnack et al., 2025] uses for adaptive item selection. Recent work has applied multidimensional IRT directly to MovieLens-1M, showing that MIRT can support interpretable latent profiles and exploration–exploitation control [Veldkamp et al., 2025].

2 The FIRE Algorithm

$$\text{FIRE}(u, i) = \alpha_u \cdot P_{\text{eff}}(u, i) + (1 - \alpha_u) \cdot \bar{I}(u, i) \tag{1}$$

$P_{\text{eff}} = (1 - \lambda_i) \cdot \sigma(a_i(\theta_u - b_i)) + \lambda_i \cdot \sigma(-\mathbf{q}_i^\top \boldsymbol{\eta})$ interpolates IRT (warm items) and an LLTM item-feature-based difficulty prior [Fischer, 1973] (low-count items).

$\bar{I} = a_i^2 \cdot P_{\text{eff}}(1 - P_{\text{eff}})/I_{\text{max}}$ is a normalized Fisher-information-shaped exploration term; for warm items where $\lambda_i = 0$, this reduces to standard 2PL Fisher information.

$\alpha_u = \sigma(\log(n_u/N_0))$ adapts from exploration to exploitation. $\lambda_i = \max(0, 1 - n_i/M_0)$ weights LLTM for low-count items.

Algorithm 1 FIRE Recommendation

Require: Training data, item features $\mathbf{Q}$, thresholds N_0, M_0

- 1: Fit IRT: $\{\theta_u, b_i, a_i\}$ via SGD
 - 2: Fit LLTM on warm items ($n_i \geq M_0$): $\boldsymbol{\eta} = (\mathbf{Q}_w^\top \mathbf{Q}_w + \lambda \mathbf{I})^{-1} \mathbf{Q}_w^\top \mathbf{b}_w$
 - 3: **for** user u , full candidate catalog $\mathcal{C}_u$ **do**
 - 4: Score, rank, return top- K
 - 5: **end for**
-

3 Experimental Setup

Data. MovieLens-1M [Harper and Konstan, 2015]: 6,040 users, 3,883 valid movies (max ID 3,952; 69 missing IDs excluded), 1,000,209 ratings. Temporal 80/20 split per user. Binary: rating ≥ 4 .

Features. 22 features from training data only (no test leakage): 18 genres (indexed by movie ID), log-popularity, average rating, rating std, year.

LLTM. Fitted on warm items only ($n_i \geq 10$). The LLTM component provides an item-feature-based difficulty prior, not a user-conditioned prediction. Reported squared correlation is in-sample.

Training. All models (IRT, SVD, FIRE) are fitted via stochastic gradient descent. For computational consistency across baselines, each SGD epoch uses a random subsample of up to 100,000 training interactions.

Baselines. IRT 1D, SVD-50, Popularity (Positive-count MostPop: items ranked by training positive-interaction count; the same scoring is used for both AUC and Top-K evaluation).

Evaluation. Full-catalog candidates (no sampling): each model scores all valid unrated items per user. 300 randomly sampled eligible users per seed. Coverage denominator: all 3,883 valid items. All results: 3 seeds, mean $\pm$ SEM. Because only three seeds are used, SEM values are descriptive rather than formal uncertainty estimates.

4 Results

Table 1: AUC on temporal held-out interactions (mean $\pm$ SEM, 3 seeds).

Method	AUC	Δ vs Pop
SVD-50	0.7823 $\pm$ 0.0000	+0.1015
IRT 1D	0.7766 $\pm$ 0.0002	+0.0958
FIRE	0.7682 $\pm$ 0.0002	+0.0874
Pop	0.6808 $\pm$ 0.0000	—

Popularity dominates Top-K accuracy (HitRate 44.7%) and absolute coverage (0.0189), consistent with Ji et al. [Ji et al., 2020]. Among the learned models, FIRE achieves the highest coverage (0.0179), 1.8 $\times$ that of IRT (0.0100), while trading accuracy for diversity: HitRate 23.0% vs IRT’s 26.3%.

Table 2: Top-10 recommendation, full-catalog evaluation (mean $\pm$ SEM, 3 seeds, 300 users/seed).

Method	Recall@10	NDCG@10	HitRate@10	Coverage
Pop	0.0560 ± 0.0019	0.1077 ± 0.0058	0.447 ± 0.013	0.0189 ± 0.0011
SVD	0.0227 ± 0.0019	0.0441 ± 0.0022	0.262 ± 0.012	0.0162 ± 0.0008
IRT	0.0230 ± 0.0013	0.0406 ± 0.0012	0.263 ± 0.007	0.0100 ± 0.0004
FIRE	0.0185 ± 0.0007	0.0346 ± 0.0011	0.230 ± 0.003	0.0179 ± 0.0005

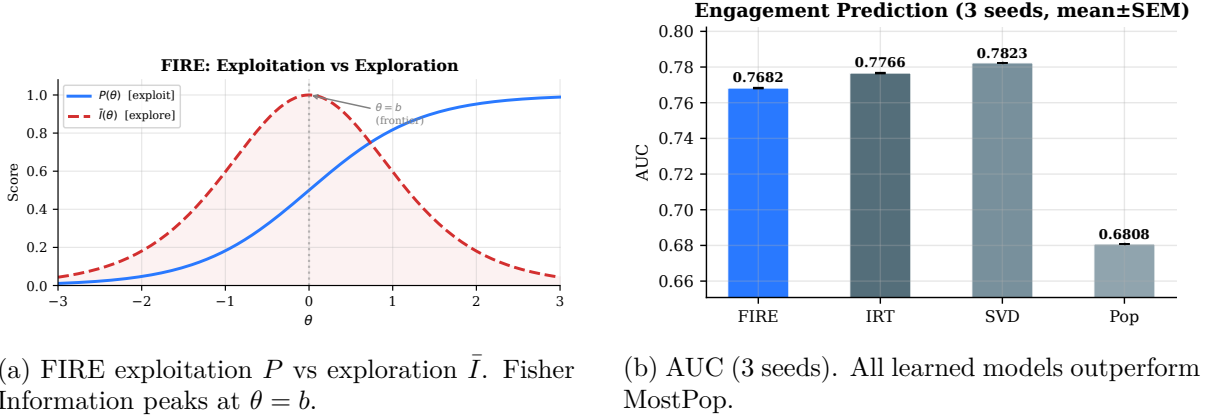


Figure 1: FIRE mechanism and engagement prediction.

4.1 LLTM Feature Decomposition

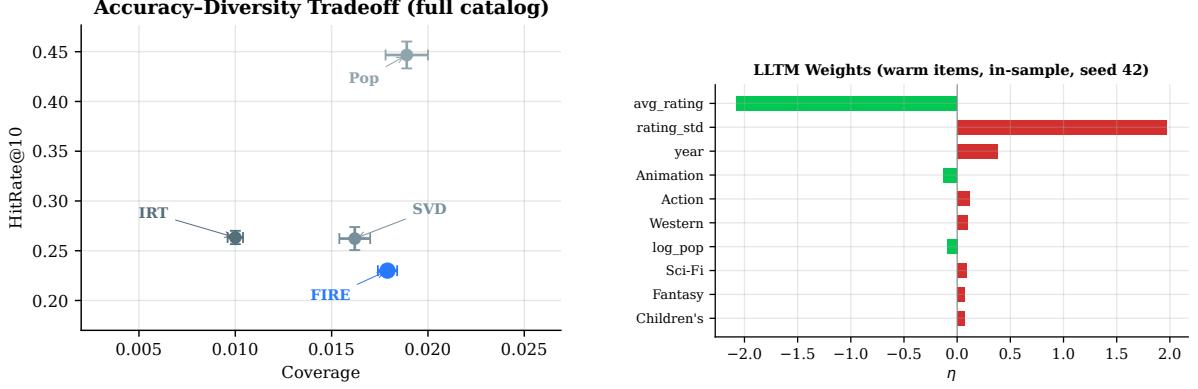
LLTM achieves an in-sample squared correlation of $\rho^2 = 0.5639 \pm 0.0023$ between 22 observable features and fitted warm-item IRT difficulty. Out-of-sample predictive validation remains future work.

4.2 Low-Count Items

On items with fewer than $M_0 = 10$ training interactions, IRT achieves AUC 0.6643, outperforming FIRE (0.6162) and Popularity (0.6012). The LLTM interpolation mechanism does not improve low-count item prediction on this dataset. True zero-shot new-item evaluation remains future work.

4.3 Explainability

FIRE provides explicit per-component score decompositions: θ_u , b_i , P_{eff} , $\bar{I}$, α_u , λ_i , and LLTM feature weights. Compared with many black-box ranking pipelines, FIRE exposes score components that can be inspected individually. This structure may support transparency-oriented documentation under frameworks such as the EU AI Act [European Union, 2024]. This paper does not make a legal compliance claim.



(a) Accuracy-diversity tradeoff. FIRE covers $1.8\times$ more items than IRT among learned models.

(b) LLTM weights (seed 42, warm items, in-sample). Average rating drives engagement ease.

Figure 2: Coverage tradeoff and LLTM feature decomposition.

5 Platform Mapping

FIRE maps onto existing architectures as a complement, not a replacement: for TikTok-style interest-graph systems, it could add an adaptive Fisher-information signal alongside static business weights; for YouTube-style two-stage systems [Covington et al., 2016], it can serve as an interpretable candidate pre-filter; and for Facebook-style discovery feeds, where AI-recommended content from unfollowed accounts constituted more than 20% of feed content as of 2023 [Meta AI, 2023], it can provide transparent routing for new content.

6 Limitations

(1) SVD-50 outperforms FIRE on AUC. (2) IRT outperforms FIRE on Recall, NDCG, and HitRate; FIRE’s advantage is coverage among learned models. (3) Low-count item prediction does not benefit from LLTM interpolation. (4) LLTM squared correlation is in-sample. (5) Popularity dominates Top-K accuracy and absolute coverage [Ji et al., 2020]. (6) Single dataset; generalization is untested. (7) SEM over 3 seeds is descriptive.

7 Conclusion

FIRE connects IRT, Fisher Information, and LLTM into a recommendation algorithm with adaptive exploration-exploitation. On MovieLens-1M with full-catalog evaluation, FIRE provides $1.8\times$ more coverage than IRT among learned models, while retaining nontrivial accuracy and offering explicit per-component score decompositions.

Code. Repository: <https://github.com/testofschool/fire-recommendation>.

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